Drill Machine



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 22 labs

Covers these lab items: Drill machine; Drill machine and torque wrench; Drill machine with drill bits of different sizes; Drill machines; Drilling Machine; Electric Drill Machine; Electrical hand drill machine; Electrical hand drill machine MII Class 1 Compliant; Power drills and drivers; Powered Hand drill

What it is used for

A drill machine makes round holes in suitable materials using a rotating drill bit. Correct bit selection, secure clamping, and controlled operation help prevent injury and damage.

Main parts

- Motor or power unit rotates the drill bit.
- Chuck holds the drill bit securely.
- Drill bit cuts the hole in the material.
- Trigger or switch controls the machine.
- Handle provides grip and control during drilling.
- Direction selector, if fitted, selects forward or reverse rotation.
- Battery or power cord supplies electrical power, depending on the model.

Before you start

1. Read the machine instructions and confirm you are trained to use the drill.
2. Wear safety goggles and suitable footwear; tie back long hair and secure loose clothing.
3. Inspect the body, chuck, bit, switch, battery, or power cord for damage.
4. Choose a drill bit suitable for the material, hole size, and machine rating.
5. Disconnect power or remove the battery before fitting or changing a bit.
6. Clamp the workpiece firmly and clear the drilling area of unnecessary objects.

How to use

1. Mark the required hole position and check the workpiece for hidden wires or pipes.
2. Select the correct bit and fit it fully into the chuck.
3. Tighten the chuck securely using its key or the keyless mechanism.
4. Remove the chuck key completely before switching on the machine.
5. Set the appropriate speed and direction according to the material and instructions.
6. Hold the drill firmly with both hands where the design allows.
7. Position the bit on the mark and keep it aligned with the surface.
8. Start the drill and apply steady, controlled pressure without forcing the bit.
9. Release the trigger before withdrawing the bit from the hole.
10. Switch off, wait for rotation to stop, and disconnect power before adjustments.

⚠ Safety

- Never wear gloves near a rotating drill bit or chuck because they can become entangled.
- Wear safety goggles and use hearing protection when required by the task.
- Keep hands away from the rotating bit and secure small workpieces with clamps.
- Do not use damaged bits, frayed cords, or a cracked machine body.
- Never leave a running drill unattended or touch the bit immediately after drilling.
- Keep liquids away from electrical parts and disconnect power before clearing jams.

Care & storage

- Clean dust and chips from the machine after switching off and disconnecting power.
- Inspect bits for wear, cracks, or bending and replace damaged bits.
- Store bits in a dry organiser and keep the drill in a dry, secure place.
- Arrange servicing for unusual noise, overheating, sparking, or damaged electrical parts.

Common mistakes to avoid

- Changing a bit while the machine remains connected to power.
- Holding the workpiece by hand instead of clamping it securely.
- Applying excessive pressure or using the wrong speed for the material.
- Leaving the chuck key in place before starting the drill.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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